

Severity Classification of Dysarthric Speech in ALS Patients

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Abstract—Neurodegenerative diseases such as amyotrophic lateral sclerosis (ALS) present major challenges in clinical monitoring and early diagnosis as current evaluation methods rely heavily on subjective human judgment. Over the period of this course, our team participated in the Speech Analysis for Neurodegenerative Diseases (SAND) challenge hosted by ICAR-CNR, the Institute for High Performance Computing and Networking within the Consiglio Nazionale delle Ricerche (CNR). This challenge was created to address the need for noninvasive, objective, and scalable biomarkers such as speech signals which could be used in early diagnosis and to monitor the progression of neurodegenerative diseases. Using machine learning data analysis techniques including classical decision-tree models and the Audio Spectrogram Transformer (AST) model developed by Gong et al. [1], our research begins to identify a potential method for accurate classification of the severity of dysarthria in patient-provided speech signals. Based on our training and testing results, we propose that future research in audio classification for neurodegenerative diseases should explore the use of AST or other transformer-based models more thoroughly, as well as work to solve the complication of unbalanced data. This research attempts to improve the diagnostic accuracy and quality of care provided to patients with ALS and other neurodegenerative diseases by classifying symptoms with machine learning techniques.

Index Terms—*Amyotrophic lateral sclerosis (ALS); dysarthria; severity classification; Audio Spectrogram Transformer (AST)*

I. INTRODUCTION

Speech signals have been proposed as a noninvasive biomarker with the potential to classify the severity of voice disorders, which would have a great impact on the early diagnosis and progression monitoring of neurodegenerative diseases. Unlike some diagnostic tools for neurodegenerative diseases such as amyotrophic lateral sclerosis (ALS), speech signals are objective and noninvasive, which means that their recording is not subject to human subjectivity and is harmless to the patient. Compared to imaging or electrophysiological measurements, speech is easy to collect, inexpensive, and well-suited for longitudinal tracking.

Patients suffering from ALS often develop a voice disorder, referred to as dysarthria, which deteriorates their quality of speech due to imprecise articulation, irregular speech rate, and lower volume. Dysarthria is currently assessed based on a perceptual rating scale called the ALSFRS-R scale, but this scale uses a clinician’s subjective assessment of patient symptoms and, as a result, can be fallible and inconsistent with low sensitivity to slight changes. Therefore, our research focused on identifying a reliable approach for correctly detecting and classifying the severity of dysarthria by analyzing the patient-provided speech signals with a trained model.

II. RELATED WORK

A substantial body of research supports the connection between acoustic parameters and motor speech disorders. Studies consistently highlight the importance of fundamental frequency (F0), jitter, shimmer, the harmonic-to-noise ratio (HNR), and various cepstral and spectral measures in quantifying deviations in vocal quality. For example, Dubbioso et al. [2] found that the standard deviation of fundamental frequency was particularly effective in distinguishing patients with dysarthria from healthy speakers, a finding that aligns with long-standing theory that pitch irregularity is a strong indicator of impaired voice control. In a similar analysis using voice recordings of vowel phonation and reading of a phrase, Rocha et al. [3] demonstrated that audio features such as pause duration and spectral bandwidth correlate strongly with severe stages of ALS, especially in tasks involving rhythmic speaking.

Machine learning techniques have been widely applied to dysarthria classification. Classical approaches frequently rely on feature selection and ensemble methods. Yeo et al. [4] combined Recursive Feature Elimination with Extra Trees Classifiers (ETC) and achieved classification accuracy above 77% for multi-class dysarthria severity. Their work shows that extracted numerical audio features still have value, especially when datasets are small or highly imbalanced. Ensemble models like Random Forests also provide robustness against noise and variability but can struggle when data contains extreme class imbalance, as is common in ALS research.

More recently, deep learning methods have demonstrated superior performance. Models based on Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and self-supervised models such as wav2vec2 and HuBERT have reached accuracies above 90% on benchmark dysarthria datasets. These models operate directly on raw audio, spectrograms, or learned embeddings, allowing them to capture complex temporal and spectral patterns that extracted audio features may miss. Ultimately, our research into audio classification methods led us to employ the Audio Spectrogram Transformer (AST) model as our final classifier, which will be described in detail further.

These studies establish a trajectory from feature-engineered approaches toward spectrogram-based deep learning methods. Our project mirrors this trajectory by performing classical machine learning experiments on extracted acoustic features, followed by the use and evaluation of an AST-based hierarchical classifier on mel spectrogram representations.

III. DATA PREPROCESSING

The dataset provided for this challenge included both quantitative patient data and audio files for each patient containing phonation and rhythm recordings of their speech. The quantitative data provided contained values for patient ID numbers, age, gender, and the ALSFRS-R scale class values assessed for each patient. The severity classifications fell in the range of values 1 to 5, where class 1 represented ALS patients with severe dysarthria, class 2 represented ALS patients with moderate dysarthria, class 3 represented ALS patients with mild dysarthria, class 4 represented ALS patients experiencing no dysarthria, and class 5 represented healthy subjects. Each patient had corresponding audio files of their speech containing vocalization of each vowel, or phonation of /a/, /e/, /i/, /o/, and /u/, as well as the reiteration of syllables, or rhythm recordings of speaking /ka/, /pa/, and /ta/.

To begin this classification task, our team started by pre-processing the audio files to gain a better understanding of the data. First, we employed a Python library for audio analysis, Librosa, to create and save mel spectrograms for each individual phonation and rhythm recording. We then explored the literature as a means of finding descriptive, numerical audio measures to be extracted. Dubbioso et al. [2] calculated audio features such as fundamental frequency (F0) mean and standard deviation, jitter, shimmer, and harmonic-to-noise ratio (HNR) to perform analysis on their audio data, which included both vowel phonation and syllable rhythm recordings in the same form as our patient audio data. Through a similar analysis, Rocha et al. [3] found strong correlations between ALSFRS-R scale scores to pause time and spectral bandwidth features from the phrase recording. In addition to these acoustic measures, we extracted cepstral peak prominence (CPP), soft phonation index (SPI), glottal to noise excitation ratios (GNE), vowel space area (VSA), formant transition rate, and articulation rate from the preprocessed spectrograms to add to the dataset, as well as some other measures which were deemed less informative. The combination of these many extracted features helped us gain clarity into the differences between the severity classifications.

Though this feature set gave an initial overview of the data and was employed in our use of classical tree-based models, our further research into models trained for audio analysis led us to a method which does not employ the use of our extracted audio features. Alternatively, we implemented the Audio Spectrogram Transformer (AST), which was first developed by Gong et al. [1] to be applied directly to an audio spectrogram. Their model is based on the Transformer architecture and most closely resembles the Vision Transformer (ViT), but differs as it can process variable-length audio inputs instead of only fixed-dimensional image inputs [1]. We selected this model over other classification methods as there were studies similar to our task, such as one performed by Bharti et al. [5], which used the AST model for binary severity classification of speech spectrograms and reported accuracies of 95.6% for speaker-dependent severity classification and 78.97% for

speaker-independent classification.

IV. ANALYSIS AND METHODOLOGY

A. Tree-Based Classification with Extracted Features

Decision Trees and Random Forests were first applied to phonation features across four binary classification tasks intended to isolate clinically meaningful distinctions in dysarthria severity. As shown in Figure 1, phonation-based balanced accuracies remained modest, generally between 0.53 and 0.63, indicating that the extracted features from phonation recordings alone do not capture the subtle acoustic differences needed to reliably separate adjacent severity levels. In the healthy versus ALS (5 vs 4to1) task, the Decision Tree achieved a balanced accuracy of 0.574, while the Random Forest reached 0.618, which is the strongest phonation result overall for the classical models. Performance declined for tasks involving more fine-grained distinctions, such as mild versus moderate and severe (3 vs 2to1) and no dysarthria versus dysarthria (4 vs 3to1), where balanced accuracies for Random Forests fell to 0.550 and 0.533, respectively. The highest raw accuracy for phonation occurred in the moderate versus severe (2 vs 1) classification, where the Random Forest reached 0.899; however, the balanced accuracy remained considerably lower (0.624), reflecting the substantial class imbalance caused by the extremely small number of severe cases.

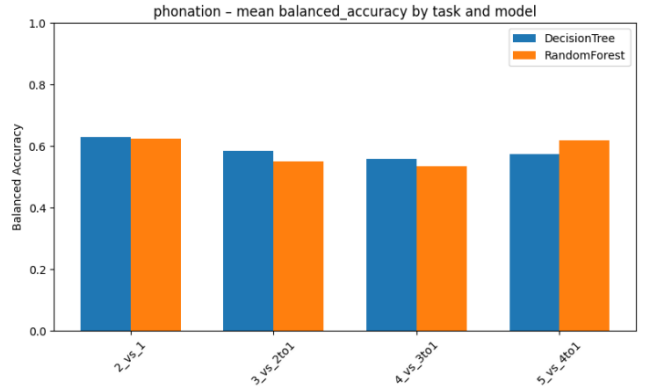


Fig. 1. Mean Balanced Accuracy by Tree-Based Model for Phonation Task

In contrast, rhythm-based features produced consistently stronger results across all tasks, with balanced accuracies ranging from 0.597 to 0.761, as illustrated in Figure 2. Random Forests again outperformed Decision Trees, achieving 0.685 balanced accuracy in the healthy versus ALS task and 0.762 in the moderate versus severe task, which was the highest among all classical feature-based evaluations. Rhythm features appear especially sensitive to the articulatory timing disruptions caused by dysarthria, making them more discriminative than steady-state phonation features.

Despite these gains, both feature sets demonstrated limitations. Even using rhythm features, model performance rarely exceeded the mid-0.70s range, and the classifiers struggled to distinguish intermediate classes such as mild compared to moderate dysarthria. These findings suggest that extracted

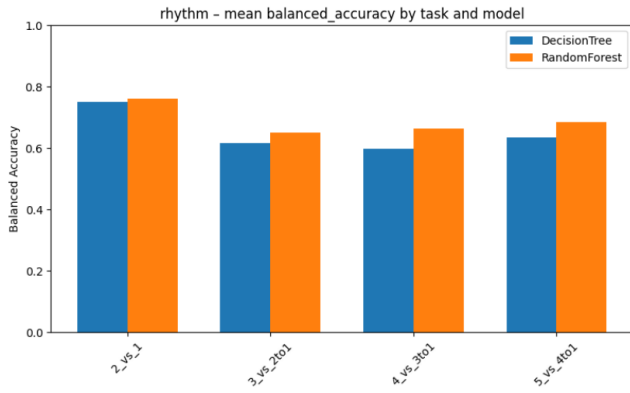


Fig. 2. Mean Balanced Accuracy by Tree-Based Model for Rhythm Task

acoustic features are insufficient for capturing the full complexity of dysarthric speech and motivated the transition to a deep-learning approach capable of learning richer representations directly from spectrograms.

B. Audio Spectrogram Transformer (AST)

Following the structure of the AST model as a guide, we preprocessed the mel spectrograms for training and testing. Each audio file was loaded at a sampling rate of 16 kHz in mono-configuration and was either truncated or padded to a fixed duration of 10.24 seconds. Files shorter than this target were right-padded with zeros. The time-series signals were then transformed into mel spectrograms using Librosa’s mel filterbank implementation with a frame size of 512 samples, hop length of 160 samples, and 128 mel-frequency bins. A Hamming window with reflection padding was applied during the short-time Fourier transform to avoid distortion. After the spectrogram extraction, normalization was performed as per the AST preprocessing protocol using AudioSet statistics. The normalized spectrograms were saved as NumPy arrays for use in our model.

To offset the effects of the unbalanced distribution of ALSFRS-R severity classification scores in the dataset as seen in Figure 3, our team segregated the model classification into three stages. In the first stage, referred to as ALS detection, the model was trained with binary classification to detect class 5, or healthy patients, labelled as 0 and the remaining classes of patients with an ALS diagnosis labelled as 1. The resulting subset of patients diagnosed with ALS was then used in the next stage, referred to as dysarthria detection. In this stage, binary classification was used again to detect patients with class 4, ALS without dysarthria, labeled 0 and the remaining three classes of patients with dysarthria labeled 1. Finally, the model was trained with multi-class classification using the final subset containing the three severity classes of ALS patients with dysarthria in the third stage, severity classification. To correct for the small amount of severe dysarthria samples, they were doubled to bring the total number of samples to 96.

The training dataset was stratified split into training, testing, and validation subsets, with a distribution of 70% for training,

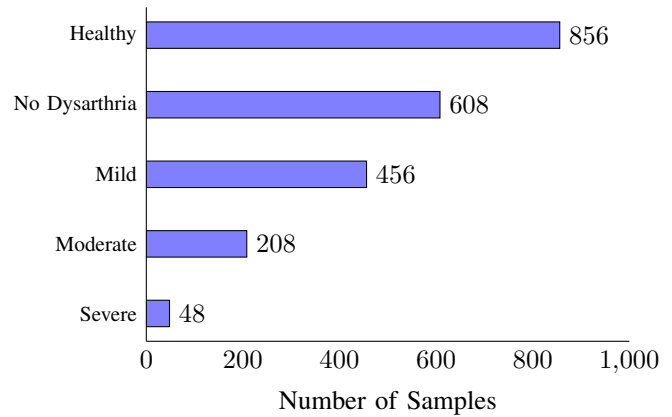


Fig. 3. Distribution of Samples Across Severity Classes

15% for testing, and 15% for validation to maintain original proportion of classes. The same stratified test set was used to evaluate all three stages as well as the overall hierarchical model performance. Since the model was trained with the use of an AST, the classifier head was trained more aggressively than the feature extractor with learning rates of 1e-3 and 5e-6, respectively, to optimize for our domain. Each model used the ReduceLROnPlateau scheduler that modified the learning rate based on how the model was performing.

The first stage, or ALS detection, was trained with 10 epochs with the best performance on validation resulting in 77.30% accuracy and an F1 score of 0.8263. Performance on the testing set resulted in an accuracy of 76.45% and an F1 score of 0.8117. Training for the second stage, also known as dysarthria detection, was performed with 10 epochs and resulted in 78.28% accuracy and an F1 score of 0.8072 for the best performance on validation. Using the testing set, the accuracy of dysarthria detection was 84.34% with an F1 score of 0.8545. Finally, the third stage, or severity classification, was trained with 15 epochs with the best performance on the validation set resulting in 88.60% accuracy and an F1 score of 0.8816. Performance on the testing set resulted in 80.70% accuracy and an F1 score of 0.7859. The overall performance of the complete model, combining the best iteration of each stage, had an accuracy of 78.05% with an F1 score of 0.7869.

V. DISCUSSION

The confusion matrix in Figure 4 reveals that the hierarchical model achieved strong overall performance with 32 out of 41 samples (78.0%) correctly classified. The model demonstrated perfect classification for severe dysarthria, though this class contained only one test sample. Most misclassifications occurred between adjacent categories in the disease progression hierarchy, suggesting the model captures the underlying continuum of dysarthria severity. The primary source of confusion was between ALS-No-Dysarthria and ALS-Mild-Dysarthria, with 2 out of 9 Mild-Dysarthria samples misclassified as No-Dysarthria. Additionally, the model struggled to distinguish between Healthy controls and ALS patients without dysarthria (2 Healthy samples as No-Dysarthria, and

1 No-Dysarthria sample as Healthy). These patterns make sense as the acoustic differences between healthy speech and ALS without dysarthria, or between no dysarthria and mild dysarthria, may be subtle. There was one Healthy sample classified as Moderate Dysarthria, which is the only distant cross-category error. The absence of no severe cross-category errors indicates that the model successfully captures major distinctions in speech impairment levels.

Actual	Predicted				
	Severe	Moderate	Mild	No Dysarthria	Healthy
Severe Dysarthria	1	0	0	0	0
Moderate Dysarthria	1	3	0	0	0
Mild Dysarthria	0	0	7	2	0
No Dysarthria	0	1	1	8	1
Healthy	0	1	0	2	13

Fig. 4. Confusion Matrix for ALS Dysarthria Classification

VI. CONCLUSION AND FUTURE WORK

To classify the severity of voice disorders such as dysarthria in speech signals collected from patients with neurodegenerative diseases, our team proposes the use of an AST model as a specific method of classifying audio spectrogram images. We applied this approach to the classification task and ALS patient dataset for the SAND challenge due to the existence of similar studies with significant accuracy in the research literature, performed by Gong et al. [1] and Bharti et al. [5]. Though we worked on extensive feature extraction, descriptive analysis, and classical machine learning models such as decision trees in addition to the AST model, we evaluated the AST model as the most effective for this classification task as it outperformed the classical approaches during training and initial testing. Our results highlight the potential of the AST model as a reliable approach to objectively classifying the severity of dysarthria on the ALSFRS-R scale, which is an essential step to improving diagnostic accuracy and patient care compared to the currently employed perceptual methods. Though the unbalanced nature of the dataset presented difficulties for this model, future research and refinement of the three-stage hierarchical approach demonstrated in our model could function as a potential solution to this issue. Advancements in classifying the severity of dysarthria could have a great impact on the quality of life and treatment of patients with neurodegenerative diseases.

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